Mooney Faces

Project Part 1 · individual project · 100 photographs and their binary transformations

What survives when shading becomes binary?



Original **Mooney**

Example 010 · source FFHQ 00041 · 1024 × 1024

Hadis Nouri

Depth Anything V2 Small + CLIP ViT-B/32

100 originals · 100 Mooneys · 2 vision systems

Faces under a controlled loss of shading

Question: do category preference, depth structure, and same-photo matching survive the Mooney transformation?

01 Choose the images

100 aligned FFHQ photographs, each supplied at 1024 × 1024.

512 candidates screened; 150 pairs visually reviewed by the assistant for facial structure and background clutter.

No model predictions used for selection.

02 Make the Mooneys

Grayscale → Gaussian blur → global brightness threshold.

Sigma 8 px; kernel 49 px. White-pixel target 22% or 28%, recorded per image.

Student viewed the review board and selected the final ten pairs.

03 Test two systems

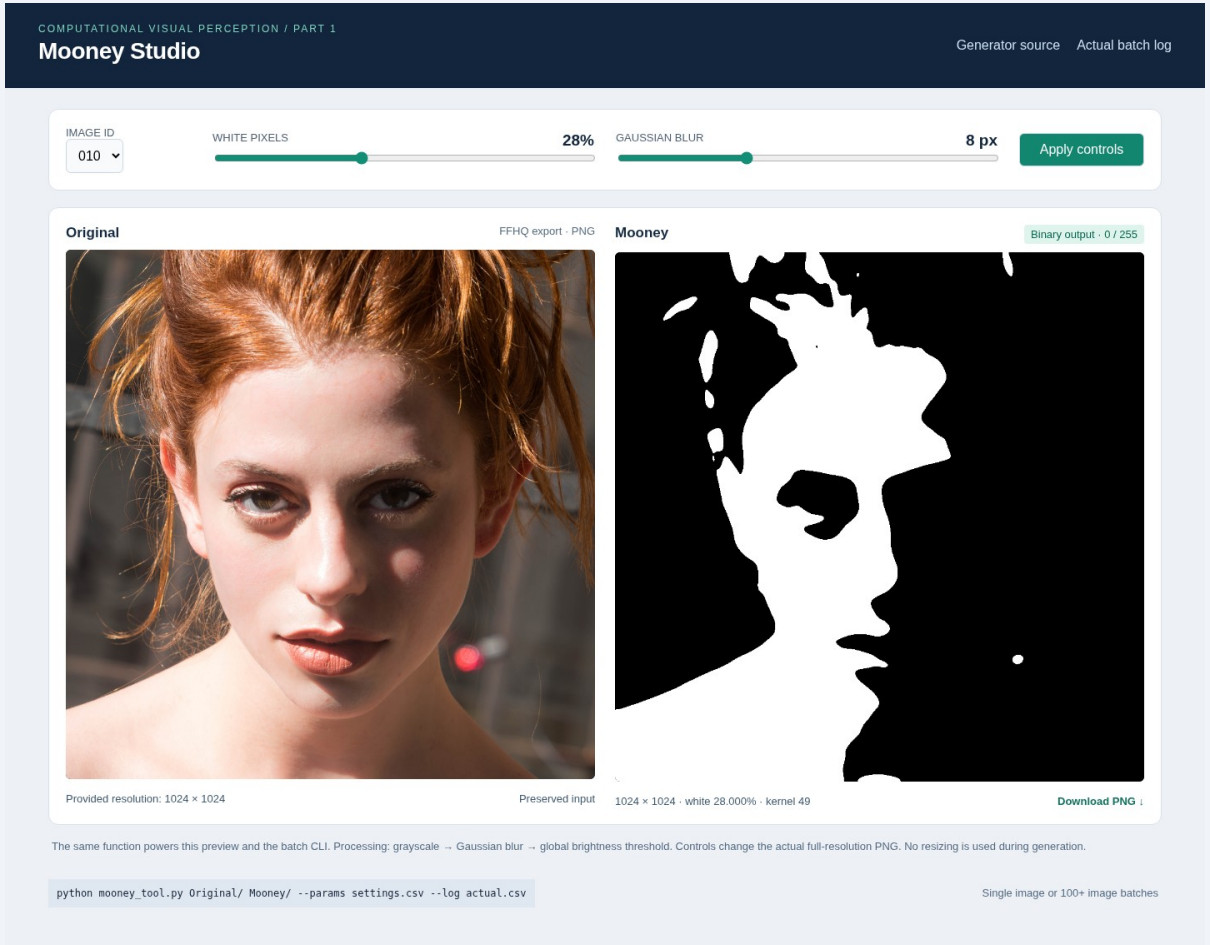
Depth Anything V2 Small: relative inverse-depth consistency.

CLIP ViT-B/32: prompt-relative face preference and embedding matching.

Both systems process all 100 originals and all 100 Mooneys.

Interactive controls, shared with the batch generator

Screenshot captured from the local GUI. Moving the sliders changes the actual full-resolution PNG.



Two independent controls

White-pixel percentage

Controls the proportion that remains white.

Blur sigma (pixels)

Controls smoothing before thresholding.

Batch CLI

Single images, folders, per-image CSV.

Verified: 101-image batch + nine parameter combinations + browser interaction.

A reproducible Mooney transformation

Actual source screenshot; the complete editable source is provided in the repository and source archive.

Actual generator source

```
def make_mooney(image, white_pct=28., sigma=8.):
    """Gaussian blur, global brightness threshold, deterministic boundary ties.

    A fractional percentage implies floor(N*p/100) white pixels. If identical
    intensities straddle the threshold, ties are ordered by a broader blur and
    then image coordinates. The log reports how many tied pixels were filled.
    """
    if not np.isfinite(white_pct) or not 0 < white_pct < 100:
        raise ValueError('white percentage must be finite and between 0 and 100')
    if not np.isfinite(sigma) or not 0 <= sigma <= 200:
        raise ValueError('sigma must be finite and between 0 and 200 pixels')
    if image is None or image.size == 0:
        raise ValueError('image is empty or unreadable')
    gray = image if image.ndim == 2 else cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    gray = gray.astype(np.float32)
    k = 2 * math.ceil(3 * sigma) + 1
    blurred = cv2.GaussianBlur(gray, (k, k), sigma) if sigma else gray
    target = int(np.floor(gray.size * white_pct / 100 + 1e-6))
    if not 0 < target < gray.size:
        raise ValueError('image is too small for the requested percentage')
    threshold = float(np.partition(blurred.ravel(), -target)[-target])
    mask = blurred > threshold
    deficit = target - int(mask.sum())
    ties = np.flatnonzero(blurred.ravel() == threshold)
    if deficit:
        broad = cv2.GaussianBlur(gray, (2 * k + 1, 2 * k + 1), max(1., 2 * sigma))
        order = np.argsort(-broad.ravel()[ties], kind='stable')
        mask.ravel()[ties[order[:deficit]]] = True
    result = mask.astype(np.uint8) * 255
    return result, {'width': gray.shape[1], 'height': gray.shape[0],
        'sigma': float(sigma), 'ksize': k, 'white_pct_target': float(white_pct),
        'white_pct_actual': float(mask.mean() * 100), 'threshold': threshold,
        'boundary_tied_pixels': len(ties), 'boundary_filled_pixels': deficit}
```

1 Grayscale in float32

2 Gaussian smoothing

$k = 2 \times \text{ceil}(3 \times \text{sigma}) + 1$

3 Global brightness threshold

requested white count = floor(N × p / 100)

4 Deterministic boundary ties

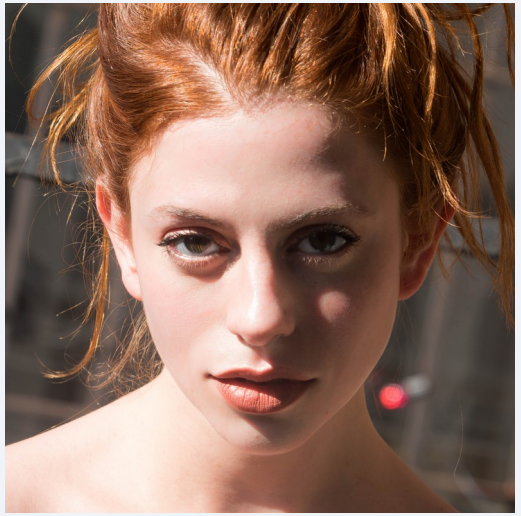
tie handling is recorded in the CSV

Generation preserves input dimensions. No image generator, edge tracing, or segmentation is used.

Blur and white percentage change different properties

One source photograph (010) is held fixed. Nine binary results are generated by the same function.





Fixed original: image 010

Rows: 16%, 28%, 40% white

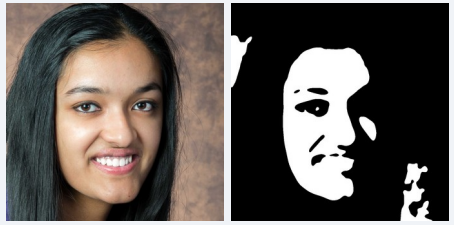
Columns: sigma 0, 8, 16 px

Final settings are logged separately for every image.

The ten manually selected pairs

Hadis Nouri viewed the review board and explicitly confirmed these ten IDs. Each pair: original on the left, Mooney on the right.

022 / white 22%



016 / white 22%



046 / white 22%



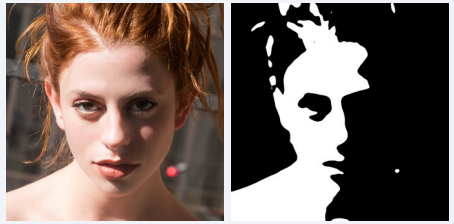
052 / white 22%



082 / white 22%



010 / white 28%



028 / white 22%



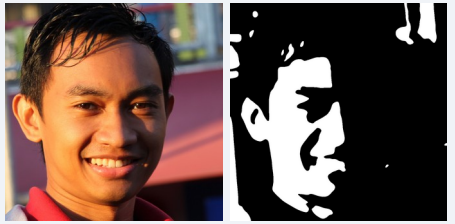
051 / white 22%



084 / white 22%



003 / white 22%



The Subset files are byte-identical copies of the corresponding full-set files.

Two pretrained vision systems, no project-specific training

One-person team: the brief requires two systems. All 200 input images are processed by each system.

01 Depth Anything V2 Small

Task: monocular relative inverse depth.

Official model: depth-anything/Depth-Anything-V2-Small-hf

Default processor; output bicubically interpolated to 1024 × 1024. Export: 16-bit normalized depth + INFERNO visualization.

02 CLIP ViT-B/32

Task: image/text similarity and visual embeddings.

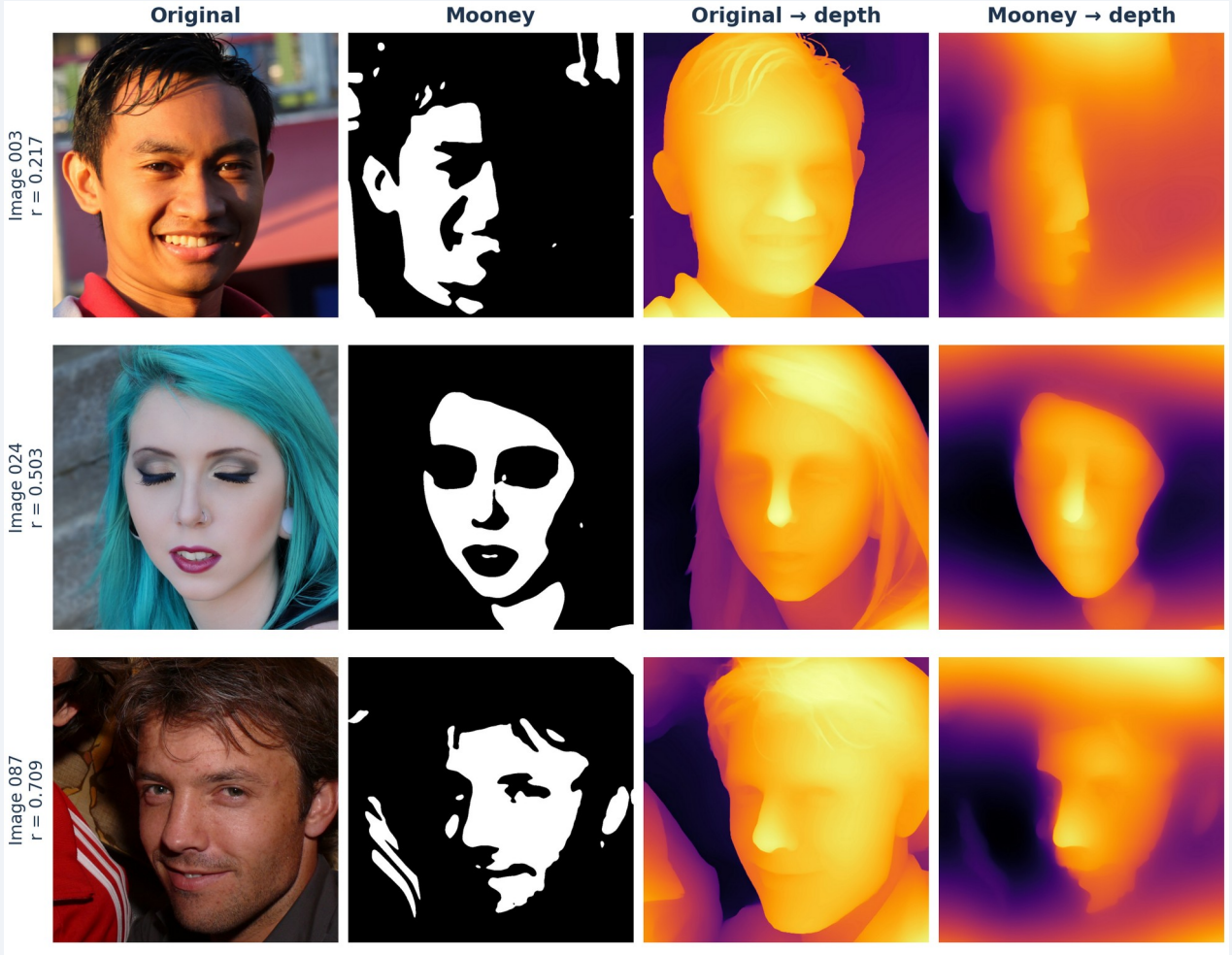
Official model: openai/clip-vit-base-patch32

Eight fixed prompts; face score sums the two face-prompt shares. Visual embeddings are unit-normalized for matching.

Official checkpoint hashes and model/config revisions are verified and stored in run_info.json.

Depth structure is partly preserved, with visible failures

Examples at the 10th, 50th, and 90th positions of the sorted paired correlations; no best-only selection.



Mean paired correlation

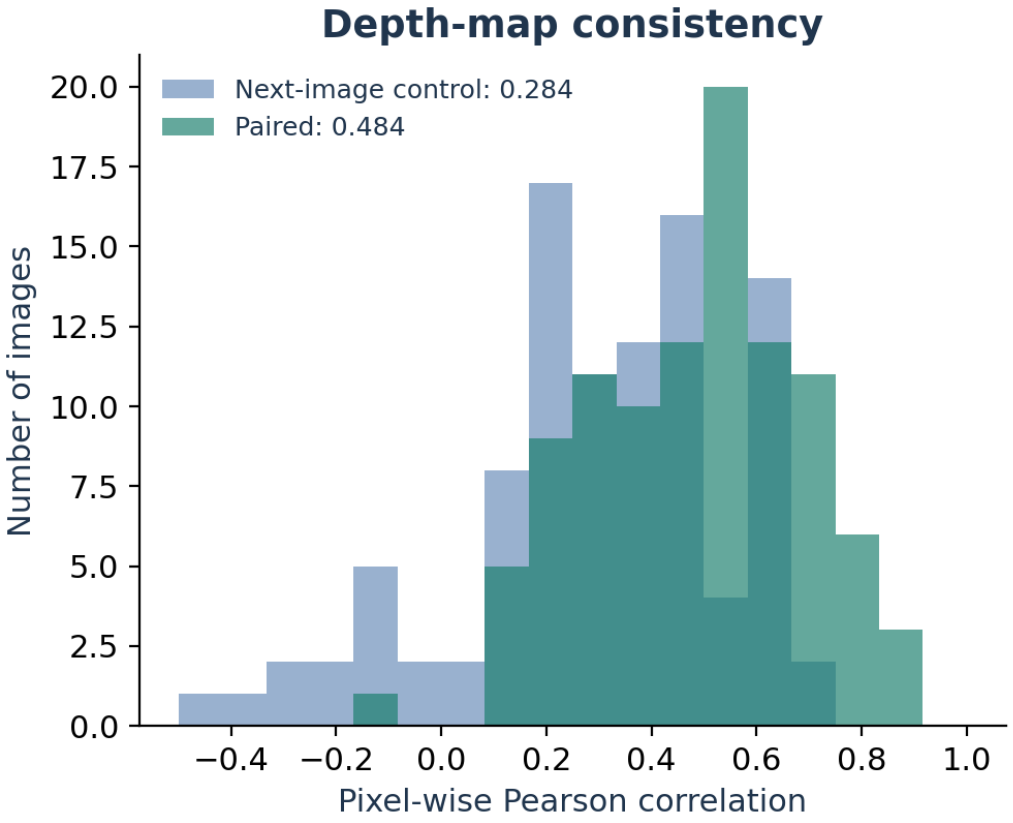
0.484

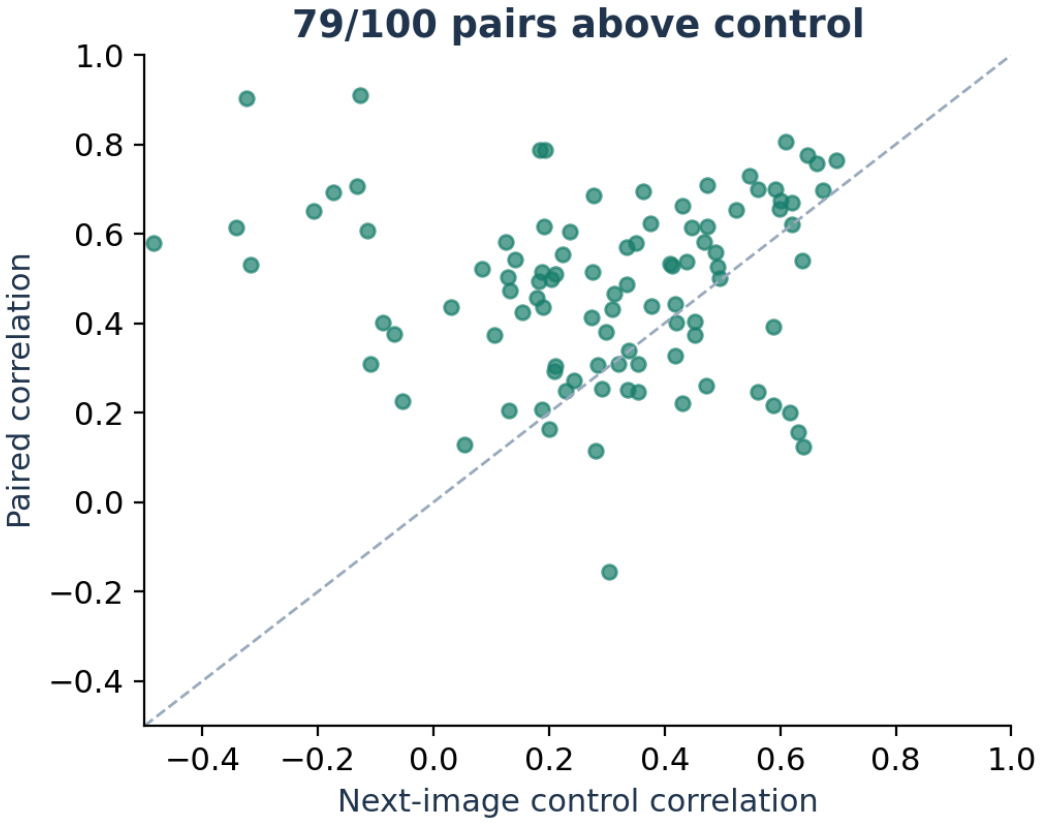
Each depth visualization is independently normalized. Brighter INFERNO colors indicate higher relative inverse depth within that image.

Color similarity across images does not establish metric depth.

Corresponding pairs exceed an unrelated-image control

Full-resolution pixel-wise Pearson correlation: original depth vs. Mooney depth; control pairs original i with Mooney i+1 (cyclic).

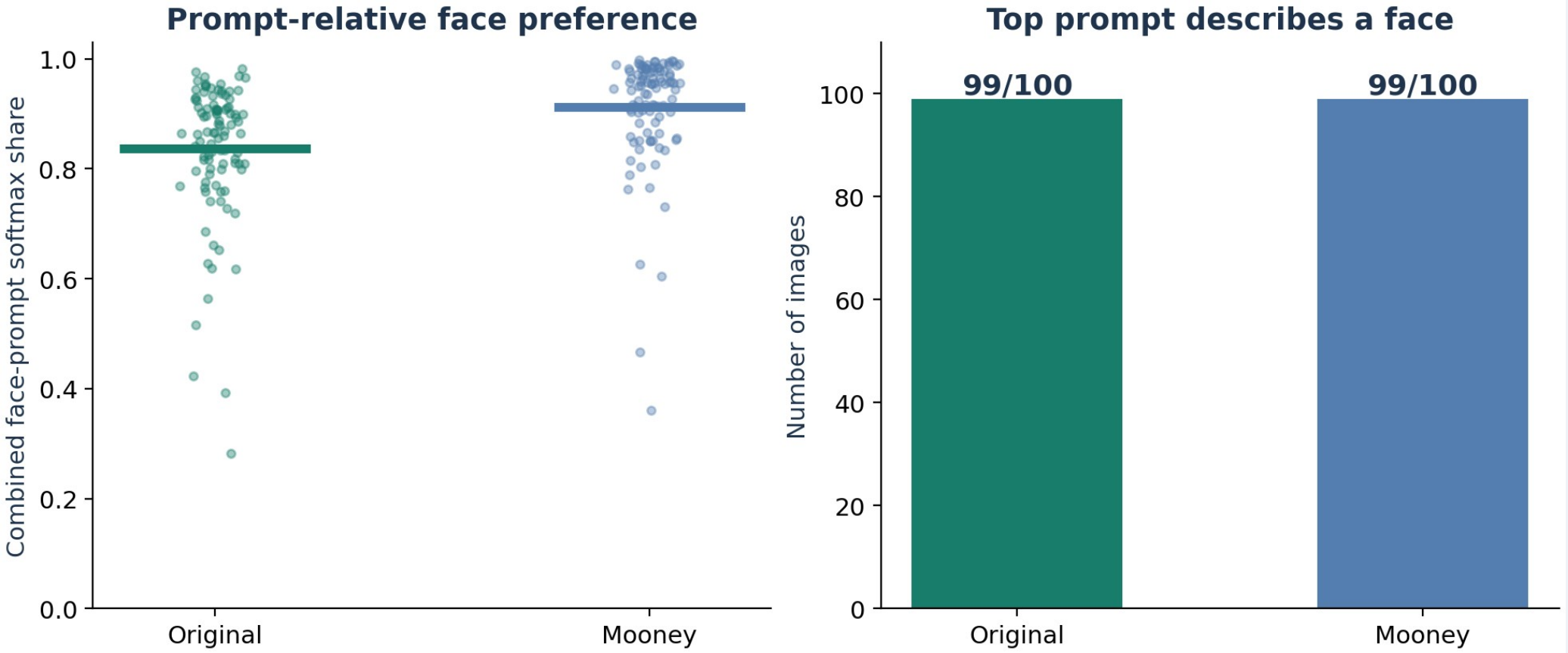




Paired mean 0.484 | control mean 0.284 | 79/100 paired maps above control

CLIP retains a strong face-prompt preference

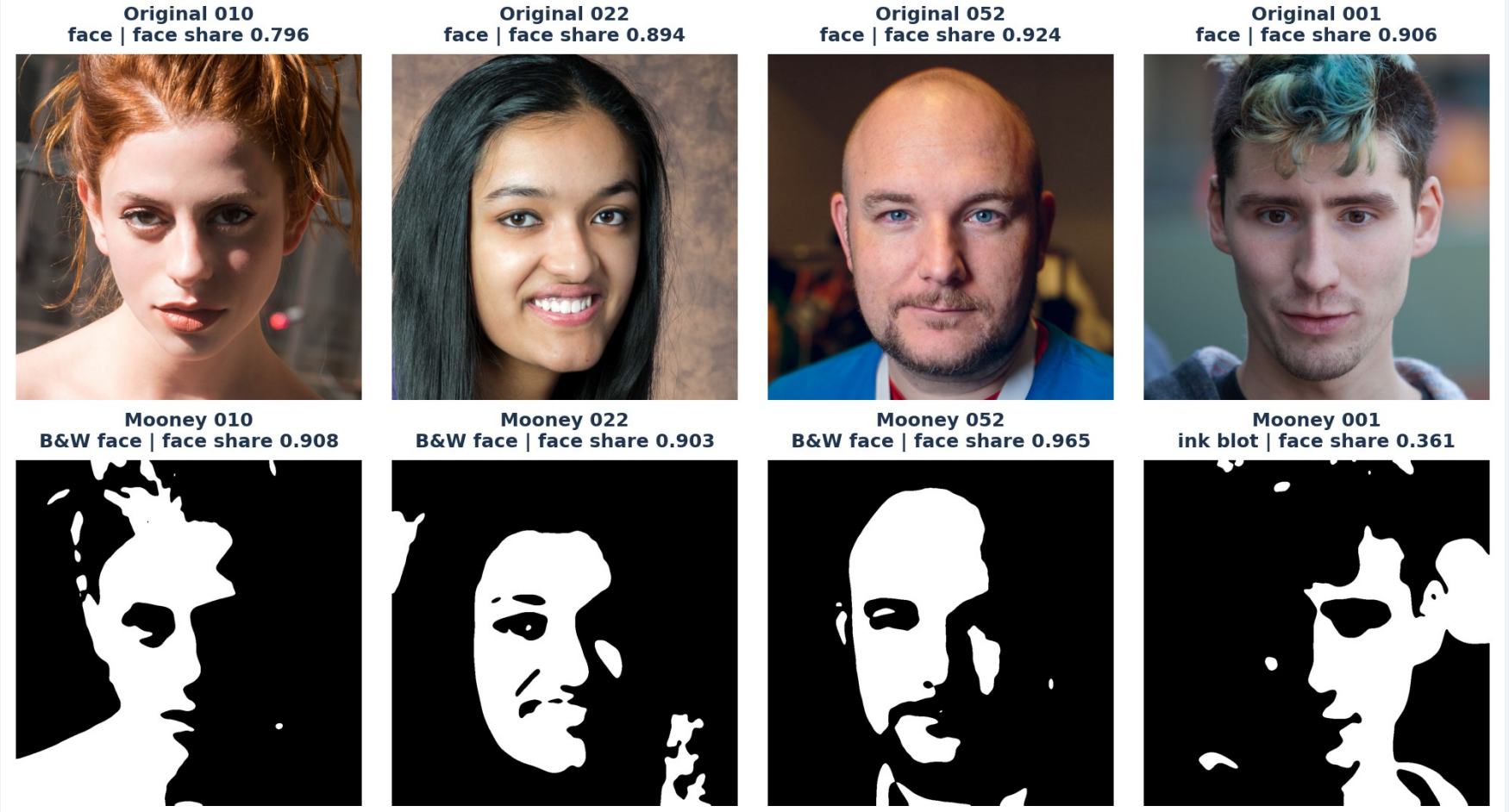
The eight-prompt comparison is fixed for originals and Mooneys. These scores are relative softmax shares, not calibrated probabilities.



Mean face-prompt share: originals 0.836; Mooneys 0.913.

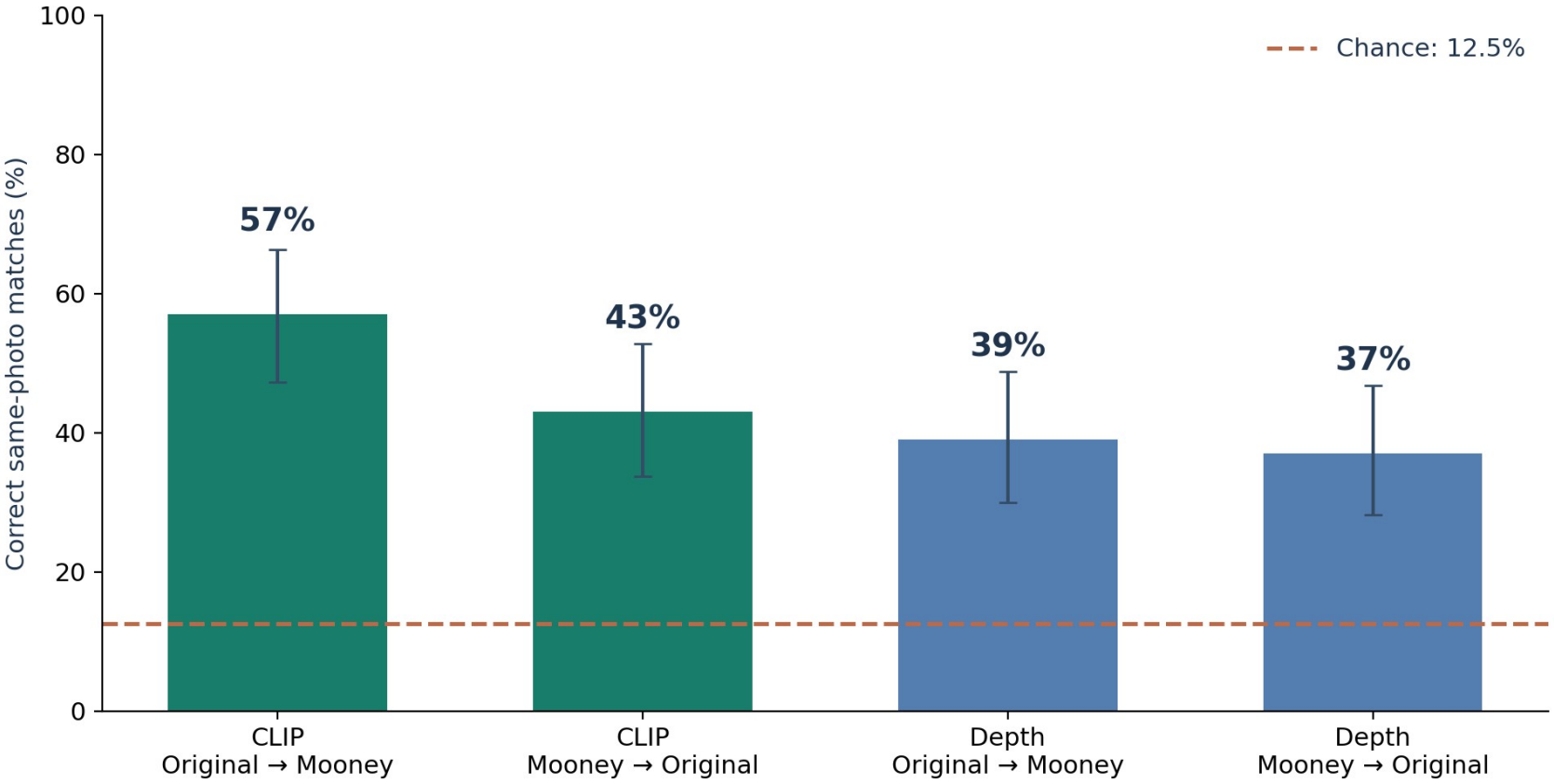
Examples include the Mooney face-prompt failure

Three subset examples plus the image whose top Mooney prompt is not a face. All eight prompt scores are saved per image.



Same-photo matching survives above the 12.5% baseline

One correct gallery image + seven distinct distractors; shuffled positions, seed 0; identical gallery design for both systems and directions.



What is being matched?

Original and Mooney views of the same photograph.

CLIP: cosine similarity of 512-D image embeddings.

Depth: correlation of maps resized to 64 × 64, centered and unit-normalized.

Error bars: Wilson 95% intervals.

Recognizing a face and recovering identity are different

Category preference is high, while matching the corresponding photograph remains substantially harder.

Paper 1: inspiration

Egger et al. compare inverse rendering with feedforward accounts on atypical faces.

This project uses their Mooney motivation and a related eight-alternative matching format.

Our same-photo test does not reproduce their cross-pose identity experiment or human comparison.

Limits of the current evidence

No true depth is available: depth-map agreement measures consistency.

The face-prompt share depends on the prompt set; it is not recognition accuracy across object categories.

Paper 2 studies fly visual motion circuits and informs later project parts; FlyVis is not one of these two systems.

The submission follows the required folder structure

Source code, provenance, parameters, embeddings and analysis tables are provided separately for reproducibility.

```
Original/                    100 PNG
Mooney/                      100 PNG
Subset/Original/              10 PNG
Subset/Mooney/                10 PNG
Result1/Original/             100 depth PNG + 100 JPEG
Result1/Mooney/               100 depth PNG + 100 JPEG
Result2/Original/             100 CLIP JSON
Result2/Mooney/               100 CLIP JSON
CompVP_Part1_Hadis_Nouri.pptx
```

Checks before packaging

100 matched image IDs

Binary Mooneys + exact white counts

Complete, finite model outputs

Ten confirmed subset pairs

PPTX and archive integrity

AI assistance is disclosed explicitly

The photographs are sourced from FFHQ. Their source credits and individual licenses are listed in the following appendix.

ChatGPT / Codex assisted with:

Reading the supplied brief and papers; writing and reviewing code; screening source images and choosing transformation settings; executing models; computing statistics; preparing figures, presentation and submission checks.

Student decision and computed evidence

Hadis Nouri reviewed the proposed subset board and selected the ten final pairs. Model outputs and numerical results were computed by the supplied programs. No generative-image model or invented human 1–5 ratings were used.

References and reproducibility records

Project brief	CompVP Project Part 1, supplied lecture slides, 26 October 2026. Tasks 1–3 and submission requirements.
Paper 1	Egger, Siegel, Arora, Soltani & Yildirim. Inverse Rendering Best Explains Face Perception Under Extreme Illuminations. Supplied PDF.
Paper 2	Lappalainen et al. Connectome-constrained networks predict neural activity across the fly visual system. Nature (2024).
Image collection	FFHQ (NVlabs). Aligned photographs from pinned mirror revision 505f94e2ecc6db64e967e8e6c8e2c2079ea0876b. Individual photo credits follow.
Depth model	Depth Anything V2 Small. Official checkpoint revision 5426e4f0f36572d16453bbda7a8389317b1bef99.
CLIP model	CLIP ViT-B/32. Official config revision 3d74acf9a28c67741b2f4f2ea7635f0aaf6f0268; verified weights and environment in run_info.json.

Photographs 001–010

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ID / FFHQ	Photographer / source	Source photograph	License
001 / 00015	Matthew Roth	Ethan one	Attribution-NonCommercial License
002 / 00018	Canadian Film Centre	CFC 25th Anniversary Celebration in LA	Attribution License
003 / 00019	Matt Ryall	Azwandi	Attribution License
004 / 00020	April Weeks	London073	Attribution License
005 / 00023	Eric Adamshick	IMG_5555	Attribution-NonCommercial License
006 / 00029	Paul L. McCord Jr.	IMG_6626_profile_large	Attribution License
007 / 00030	John Cathey-Roberts	Happy PJ-3.jpg	Attribution-NonCommercial License
008 / 00032	Mike W. Miller	IMG_2107	Attribution-NonCommercial License
009 / 00034	Aberdeen Proving Ground	Garrison Change of Responsibility, June 6, 2018	Attribution License
010 / 00041	Chris Goldberg	DSC00390.jpg	Attribution-NonCommercial License

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Photographs 011–020

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011 / 00043	KRWonders	20170715_131928__KRW1317_FullKR	Attribution License
012 / 00045	John Carkeet	A European Honeymoon: Day 4 (Rhine River and Co...	Attribution License
013 / 00058	Brandon	Matteo, 105mm test	Attribution-NonCommercial License
014 / 00079	Montclair Film	NEG_2950	Attribution License
015 / 00080	karlhans	Maria Cacao	Attribution-NonCommercial License
016 / 00091	Dani Figueiredo	Me and Felicia	Attribution-NonCommercial License
017 / 00099	Gideon	Miyuki	Attribution License
018 / 00117	Tuomas Puikkonen	DSC_0562	Attribution-NonCommercial License
019 / 00121	Presidencia El Salvador	Lanzamiento de la Política de la Persona Adulta...	Public Domain Dedication (CC0)
020 / 00123	Tyler Johnson	_MG_2972	Attribution License

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021 / 00136	snot_posse	i_dont_need_a_bike-7-2	Attribution-NonCommercial License
022 / 00149	Still Vision	DSC_9156	Attribution License
023 / 00151	Hamner_Fotos	IMG_6022_1	Attribution License
024 / 00158	Will Mendonça	In love with Blue Hair ❤️	Attribution-NonCommercial License
025 / 00159	Jean-Simon Asselin	DSC_9035_LR	Attribution License
026 / 00167	Richard Pyrker	DJ Spooky at Elevate	Attribution-NonCommercial License
027 / 00171	Jo Chou	LIEITKScottSchruteWorkshop-82	Attribution-NonCommercial License
028 / 00174	Julio Lopes	DSC_0881	Attribution License
029 / 00177	USAG Livorno PAO	DSC_1183	Attribution-NonCommercial License
030 / 00192	mcsfphotos	KW8_4208_k1	Attribution-NonCommercial License

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031 / 00205	avivsela3	IMG_0243	Attribution-NonCommercial License
032 / 00207	cd_1940	20081129-115543-00170	Attribution-NonCommercial License
033 / 00209	Guide Live	Blockparty (303 of 454)	Public Domain Mark
034 / 00213	O'Reilly Internal	170511_084627-5D3_0161	Attribution-NonCommercial License
035 / 00214	Joe King	Sly Tyler	Attribution-NonCommercial License
036 / 00215	Kerry J	Me and Kim	Attribution-NonCommercial License
037 / 00217	Lori Thantos	IMG_1681	Attribution-NonCommercial License
038 / 00219	Josh	them again	Attribution License
039 / 00220	Jedimentat44	2017-10-25_04-31-28	Attribution License
040 / 00230	The.Rohit	IMG_1629	Attribution-NonCommercial License

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041 / 00231	Colações de Grau UFG	Universidade Federal de Goiás	Attribution-NonCommercial License
042 / 00234	Emily Allen	Getting saucy at a club	Attribution License
043 / 00235	ShashiBellamkonda	Shashi-Bellamkonda2008-06-11_12	Attribution License
044 / 00236	Gabriela Cabezas	Belén	Attribution-NonCommercial License
045 / 00237	Lori Thantos	IMG_3156	Attribution-NonCommercial License
046 / 00255	RISE	SD4_9121	Attribution License
047 / 00258	Chris Lewis	Happy Professor	Attribution-NonCommercial License
048 / 00259	Rogelio A. Galaviz C.	Keith Richards	Attribution-NonCommercial License
049 / 00268	Tuomas Puikkonen	DSC_5197	Attribution-NonCommercial License
050 / 00276	Bundestagsfraktion Bündnis 90...	Frithjof Schmidt MdB	Attribution License

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Photographs 051–060

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051 / 00278	peiflickr	prSHIRE101507-135	Attribution-NonCommercial License
052 / 00281	Alex "Khaki" Vance	20131102-B57A7853	Attribution License
053 / 00282	Marius Strom	Diann at Cape Flattery	Attribution-NonCommercial License
054 / 00286	Jamwhy	1S6B3466.jpg	Attribution-NonCommercial License
055 / 00289	Rokas Andrulaitis	Blogoevgrad (291 of 540)	Public Domain Mark
056 / 00290	U.S. Department of State	Secretary Kerry Poses With Singer and Fashion I...	United States Government Work
057 / 00295	Lori Thantos	IMG_0664	Attribution-NonCommercial License
058 / 00297	Pawel Loj	first communion	Attribution License
059 / 00298	Vince Crabeo	J.H.Edits-53	Attribution License
060 / 00301	vgm8383	IMG_1591	Attribution-NonCommercial License

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061 / 00302	Charlie Cowins	_COW8886	Attribution License
062 / 00316	University of the Fraser Vall…	Student_Research_Excellence_Awards_2017-60	Attribution License
063 / 00318	WRI Ross Center for Sustainab…	3G0A7450	Attribution License
064 / 00325	Ariana Reyes	DSC_0648	Attribution License
065 / 00330	Nancetalk	IMG_8319	Attribution License
066 / 00334	ScruffoftheNeck	IMG_5628	Attribution-NonCommercial License
067 / 00344	Carlos Lopez Echeto Marrero	DSC_20100307_010527	Attribution License
068 / 00346	ShashiBellamkonda	Gnomedex Seattle Robert Scoble and ShashiB	Attribution License
069 / 00350	Daniele Vsilva	IMG_6767	Attribution-NonCommercial License
070 / 00351	Russell Mondy	DSC_9106	Attribution-NonCommercial License

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071 / 00354	Lori Thantos	IMG_3650	Attribution-NonCommercial License
072 / 00357	Mohd Fazlin Mohd Effendy Ooi	Zoo Johor	Attribution License
073 / 00368	Jason	Kathryn + Lisa	Attribution-NonCommercial License
074 / 00370	David Malouf	Original source photograph	Attribution-NonCommercial License
075 / 00389	Live Congress	281_III_CongresoLibroElectrónico_foto-javibroto	Attribution License
076 / 00392	Enrico	Laurea - Valerio 2	Attribution License
077 / 00399	U.S. Department of State	Acting Secretary Sullivan Participates in the G...	United States Government Work
078 / 00400	Coach Bolton and his LAHS pho...	0L9A2672	Public Domain Mark
079 / 00407	John Benson	Farmers Market	Attribution License
080 / 00426	nadir carollo	senza parole	Attribution License

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Photographs 081–090

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081 / 00428	PSM-Entesa	Biel Barceló i Milita	Attribution-NonCommercial License
082 / 00431	Mike Brand	Me at Eurovision 2008	Attribution License
083 / 00433	TÀI FOTO	_AIA9321	Public Domain Mark
084 / 00436	Aireal Robbins	DSC_0232	Attribution License
085 / 00437	__skwrl__	sKizoFrenia Exploaration #5 09-7-3 #20	Attribution-NonCommercial License
086 / 00439	South Lanarkshire Council	John Cairney - Ward 10	Attribution-NonCommercial License
087 / 00442	Todd L. Gilbert	P1020444	Attribution-NonCommercial License
088 / 00445	Remember-The-Ten-Run	IMG_2071_p-(ZF-3864-42778-1-614)	Attribution License
089 / 00450	Brady MacDonald	Stylin'	Attribution-NonCommercial License
090 / 00453	shelgerard	Super Bowl XLIV - Saints vs. Colts	Attribution-NonCommercial License

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Photographs 091–100

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ID / FFHQ	Photographer / source	Source photograph	License
091 / 00455	Laurel White	DSC00012	Attribution-NonCommercial License
092 / 00458	Malcolm Slaney	2017-10-31-17-19-06-2.jpg	Attribution License
093 / 00474	AbigailT's Photography	Whit16-1	Attribution License
094 / 00475	jade anderson	_J081874	Attribution License
095 / 00484	Aberdeen Proving Ground	20170223_GAPG_028	Attribution License
096 / 00493	Laura	Original source photograph	Attribution-NonCommercial License
097 / 00500	RISE	SD5_3909	Attribution License
098 / 00504	Senterpartiet (Sp)	Bjørn Nesse Hunderi	Attribution License
099 / 00505	Katie Cauthen	bloggermay23-2147	Attribution License
100 / 00506	osseous	March 19, 2016	Attribution License

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